

# Sudip Sharma

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## EDUCATION

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| <b>Morgan State University</b>                                                                       | Baltimore, MD       |
| Doctor of Philosophy (Ph.D.) in Advanced Computing                                                   | Expected May 2029   |
| Master of Science in Advanced Computing                                                              | Aug 2024 – May 2026 |
| <b>Relevant Coursework:</b> Design and Analysis of Algorithms, Software Engineering, Cloud Computing |                     |

## TECHNICAL SKILLS

**Programming:** Python, TypeScript, Swift | **Computer Science:** Data Structures, Algorithms, Software Design, Object-Oriented Programming  
**Backend & APIs:** Node.js, NestJS, Express, GraphQL, REST APIs, WebSockets, JWT | **Cloud:** AWS (EC2, S3, VPC, SageMaker)  
**Machine Learning & Data:** TensorFlow, NumPy, Pandas, Scikit-learn, Fairlearn, Matplotlib, Seaborn | **Databases:** MongoDB, MySQL  
**Application Development:** SwiftUI, UIKit, React

## PROFESSIONAL EXPERIENCE

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| <b>Center for Equitable Artificial Intelligence and Machine Learning Systems (CEAMLS)</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Summer 2025 & 2026 |
| Graduate Mentor, Morgan State University                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | Baltimore, MD      |
| <ul style="list-style-type: none"><li>Led interdisciplinary undergraduate research teams developing Python and TensorFlow healthcare AI systems, providing technical direction on model design, experimentation, code quality, and research methodology.</li><li>Built multimodal ECG classification pipelines using 1D-CNN, Transformer, and 2D-CNN architectures, achieving 90%+ accuracy and improving ensemble performance by 14% through time-, frequency-, and time-frequency features.</li><li>Automated preprocessing, training, and evaluation workflows with Python, NumPy, and TensorFlow; optimized GPU data pipelines to reduce training time by 30% and implemented fairness, Grad-CAM, and robustness analyses.</li></ul> |                    |

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| <b>EB Pearls</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Jul 2023 – Aug 2024 |
| Software Engineer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | Lalitpur, Nepal     |
| <ul style="list-style-type: none"><li>Designed, tested, deployed, and maintained secure authentication flows (login, OTP, JWT) in SwiftUI, improving login success rates by 40% and user satisfaction by 63%.</li><li>Developed TypeScript and NestJS GraphQL APIs and optimized data queries, improving retrieval speed by 37% across mobile and administrative applications.</li><li>Integrated the Google Calendar API for 1K+ active users, improving scheduling efficiency and engagement through reliable event synchronization.</li><li>Collaborated with product and engineering teams on debugging, code reviews, regression testing, and releases, reducing production bug reports by 25%.</li></ul> |                     |

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| <b>EB Pearls</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Apr 2023 – Jul 2023 |
| Software Engineering Intern                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | Lalitpur, Nepal     |
| <ul style="list-style-type: none"><li>Developed and maintained iOS applications in Swift using SwiftUI, UIKit, and MVVM architecture, increasing in-app session length by 18% through improved navigation and interactions.</li><li>Built and optimized scalable REST and GraphQL backend services with Node.js and Express, improving API response time by 22% through load balancing and query optimization.</li><li>Implemented real-time notification workflows with WebSockets, increasing engagement with core product features by 13%.</li></ul> |                     |

## PROJECTS & LEADERSHIP

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| <b>Mastercard Center for Inclusive Growth</b>                                                                                                                                                                                                                                                                                                        | Atlanta, GA |
| AUC Data Science Initiative                                                                                                                                                                                                                                                                                                                          |             |
| <ul style="list-style-type: none"><li>Placed 3rd by building Scikit-learn models with Python and Pandas to analyze financial inclusion disparities across demographic groups.</li><li>Evaluated model results and visualized findings with Matplotlib and Seaborn, communicating actionable patterns in digital banking and credit access.</li></ul> |             |

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| <b>IT MEET 2022</b>                                                                                                                                                                          | Dhulikhel, Nepal |
| Non-tech Events Lead                                                                                                                                                                         |                  |
| <ul style="list-style-type: none"><li>Led cross-functional planning and execution for 5+ university events, coordinating stakeholders and increasing student participation by 21%.</li></ul> |                  |

## AWARDS

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| <b>Presidential Medal for Service and Excellence</b>                                                                                                                                          | Feb 16, 2026   |
| STAR Scholars Network                                                                                                                                                                         | Washington, DC |
| <ul style="list-style-type: none"><li>Recognized for sustained leadership and organizational impact as Media Specialist at the STAR Scholars Network's 8th Anniversary Celebration.</li></ul> |                |

## PUBLICATIONS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |  |
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| <ul style="list-style-type: none"><li>Trust and Transparency: Making AI Explainable in Healthcare, Springer Nature Switzerland (International)<br/><i>Accepted (2026 Pending Publication)</i></li><li>Interpretable Time-Series Deep Learning for Monitoring Disease Progression in Chest Radiography, IEEE eIT 2026<br/><i>Accepted (2026 Pending Publication)</i></li><li>Cross-Domain Generalization and Fairness in Skin Lesion Classification: An Empirical Study of Architectural Robustness under Human-Centric Constraints, IEEE SMC 2026<br/><i>Accepted (2026 Pending Publication)</i></li><li>Predictive Stability Versus Fairness Instability in Multi-Label ECG Classification: A Multi-Seed, Cross-Institutional Analysis, IEEE JBHI 2026<br/><i>In Progress</i></li><li>CITE-ODE: Continuous-Time Evidential Modeling for Selective Prediction under Structured Missingness in ICU Time Series, IEEE JBHI 2026<br/><i>In Progress</i></li></ul> |  |
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